

Courier Delivery Analytics

Operational and financial analysis of bicycle deliveries

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- **Tools:** Python, R, Power BI
- **Dataset:** 451 real-world bicycle courier deliveries collected in Cracow, Poland.

Project objective

The objective of this project was to analyze operational and financial aspects of urban bicycle courier deliveries.

The analysis focused on:

- delivery efficiency,
- earnings patterns,
- weather impact,
- operational behavior,
- delivery duration prediction using machine learning.

Dataset overview

Dataset characteristics:

- 451 completed deliveries
- 44 working days
- Location: Cracow, Poland
- Period: December 2025 - April 2026

Data sources:

- Glovo Rider application
- Archival weather data
- Manual operational observations

Dataset overview

Key variables:

- earnings
- tips
- delivery duration
- district
- weather conditions
- multiplier factor
- order hour

Data preparation

Feature engineering- created analytical variables:

- duration_minutes
- is_weekend
- bad_weather_flag
- hour

Cleaning Process:

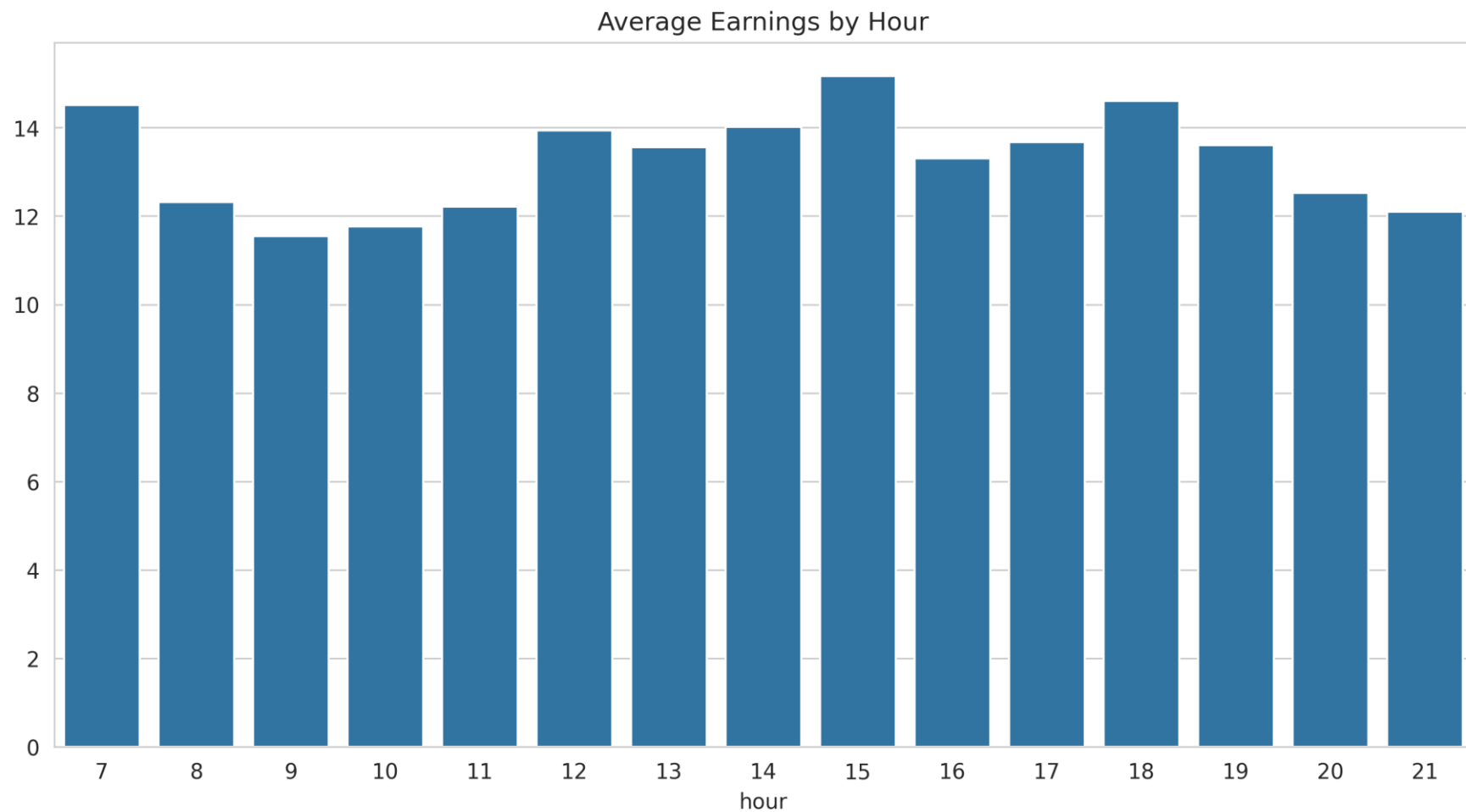
- standardized time variables
- transformed categorical features
- prepared data for machine learning models

Exploratory Data Analysis

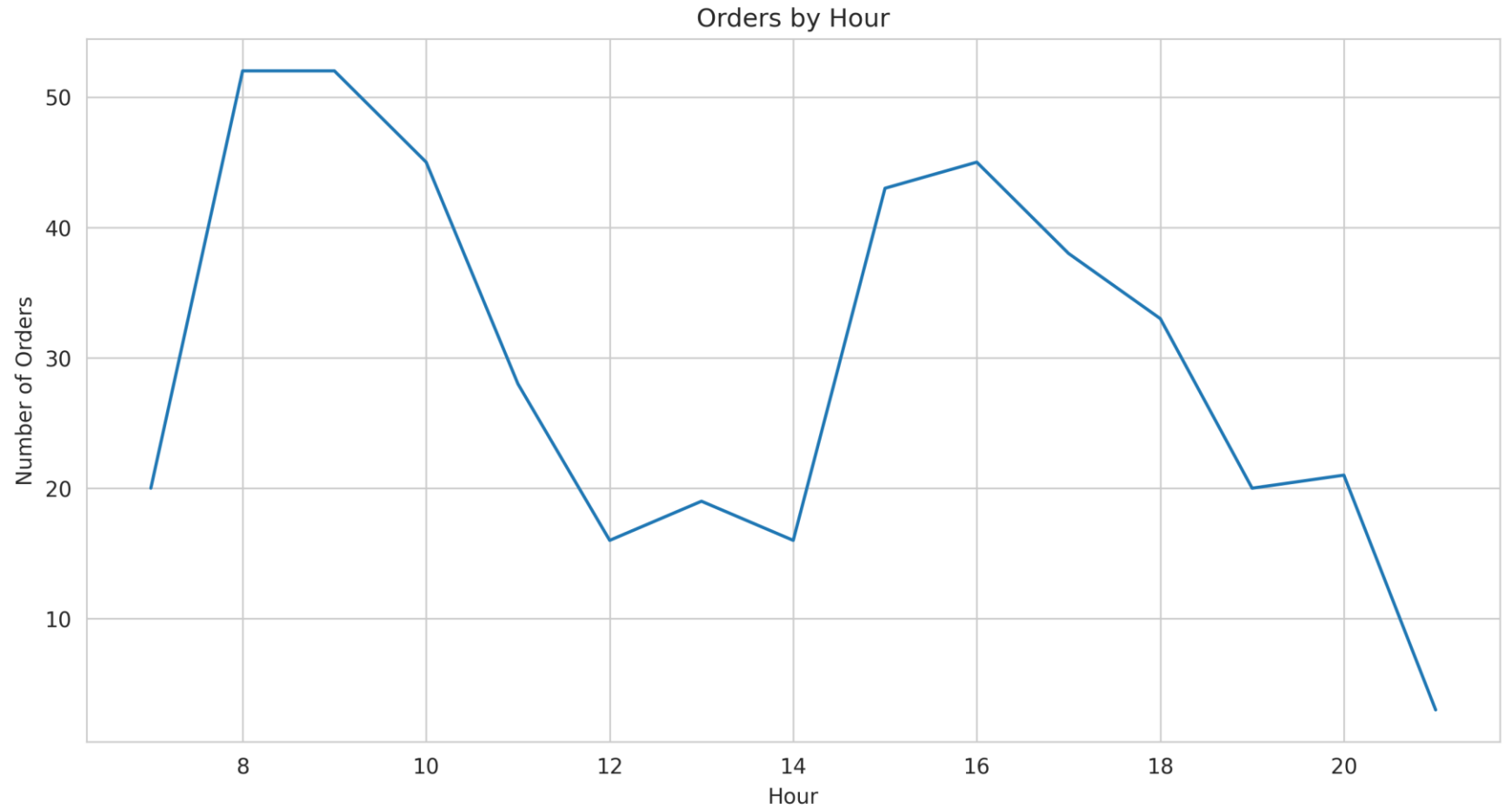
Key questions:

- When are earnings the highest?
- How does weather affect deliveries?
- Which districts are the most operationally efficient?
- How does demand change throughout the day?

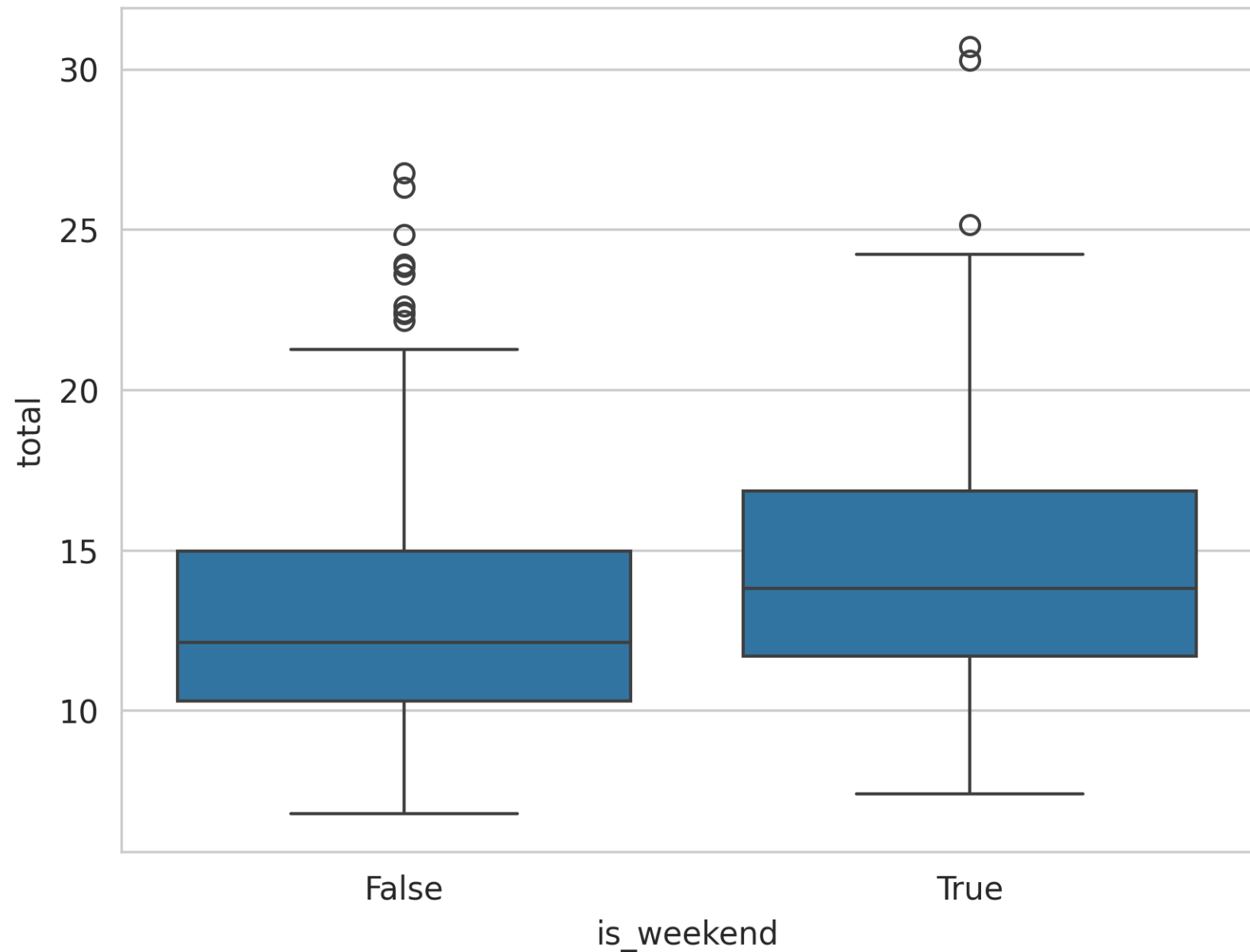
Earnings by hour (per order)



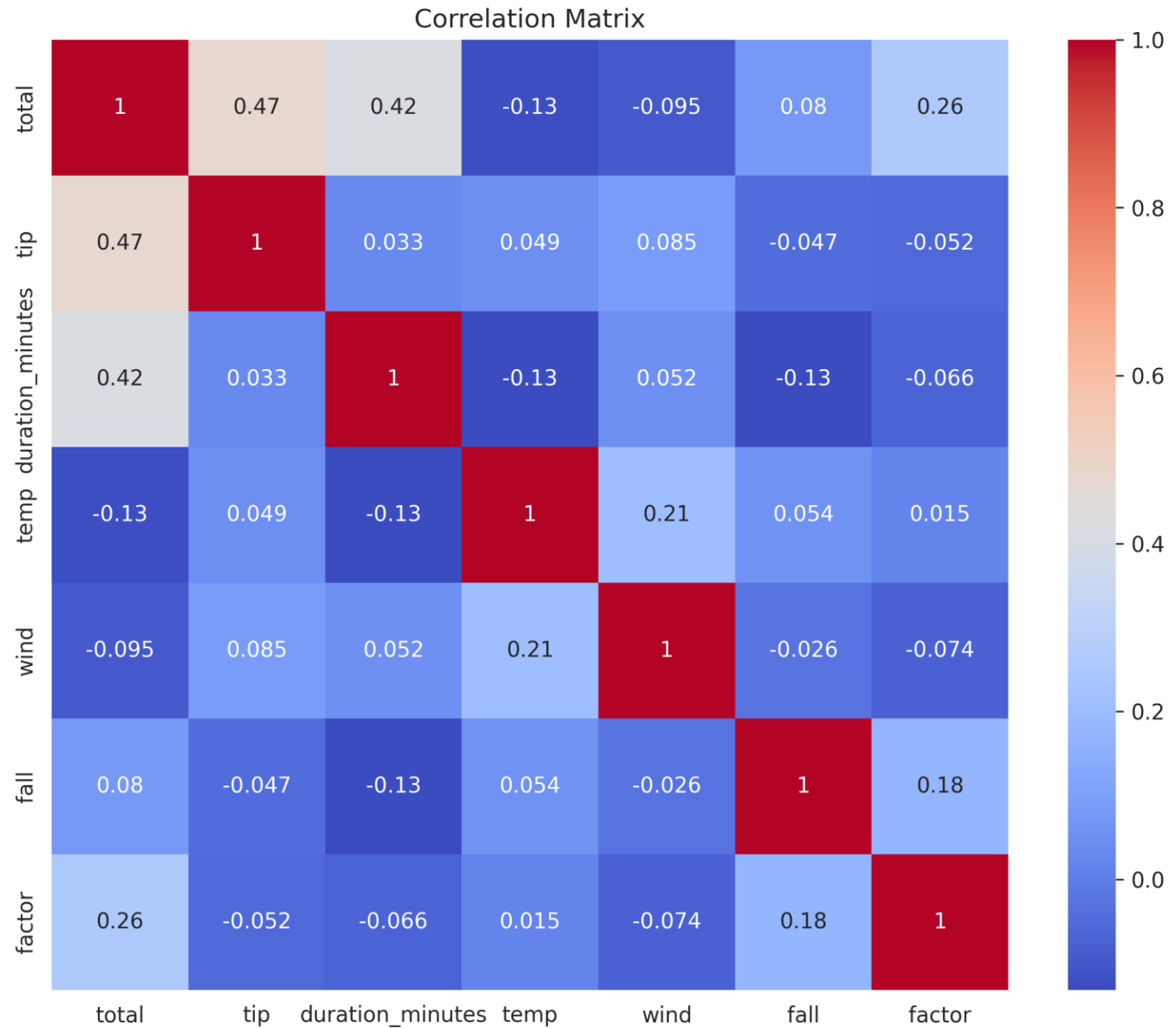
Orders by hour



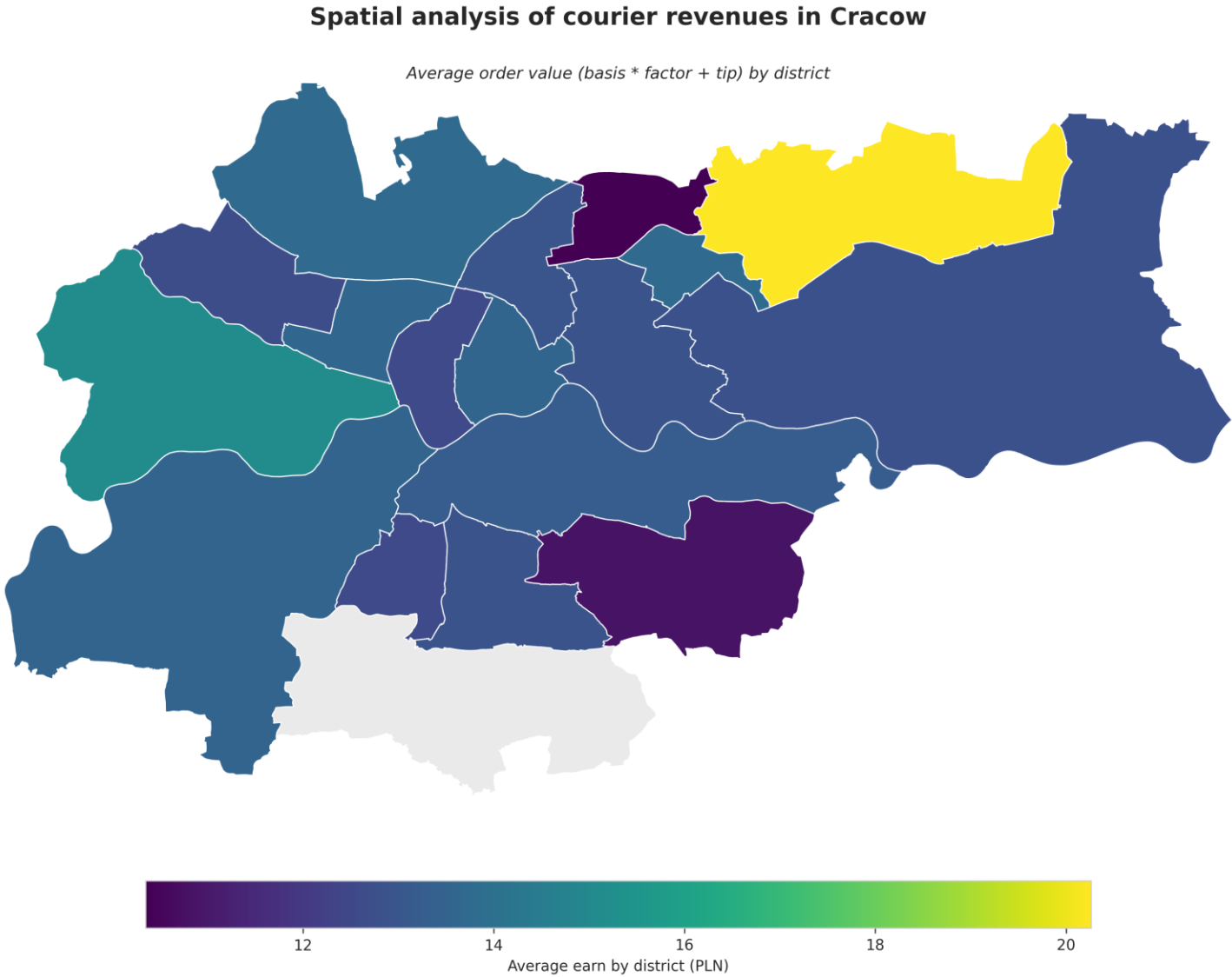
Weekend vs Working weekdays (earnings per order)



Correlation heatmap



Earnings by district



Operational insights

Key findings:

- Evening hours generated the highest earnings.
- Delivery demand increased significantly during afternoon and evening hours.
- Weekend shifts produced higher average earnings.
- Adverse weather conditions increased delivery duration.

Statistical analysis

Methods used:

- ANOVA
- Linear Regression
- Correlation Analysis

Main results:

- Temperature significantly influenced delivery duration.
- Weather conditions affected operational efficiency.
- Operational variability remained relatively high.

Statistical analysis

Regression coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	31.8388	2.5037	12.717	< 2e-16 ***
wind	0.6132	0.4117	1.489	0.13707
temp	-0.2930	0.1007	-2.908	0.00382 **
fall	-3.8449	1.5471	-2.485	0.01332 *
factor	-0.8093	1.1186	-0.723	0.46976
is_weekend TRUE	-0.7047	1.3305	-0.530	0.59662

Machine Learning

Goal:

- Predict delivery duration using operational and weather-related variables.

Models used:

- Linear Regression
- Random Forest

Evaluation metrics:

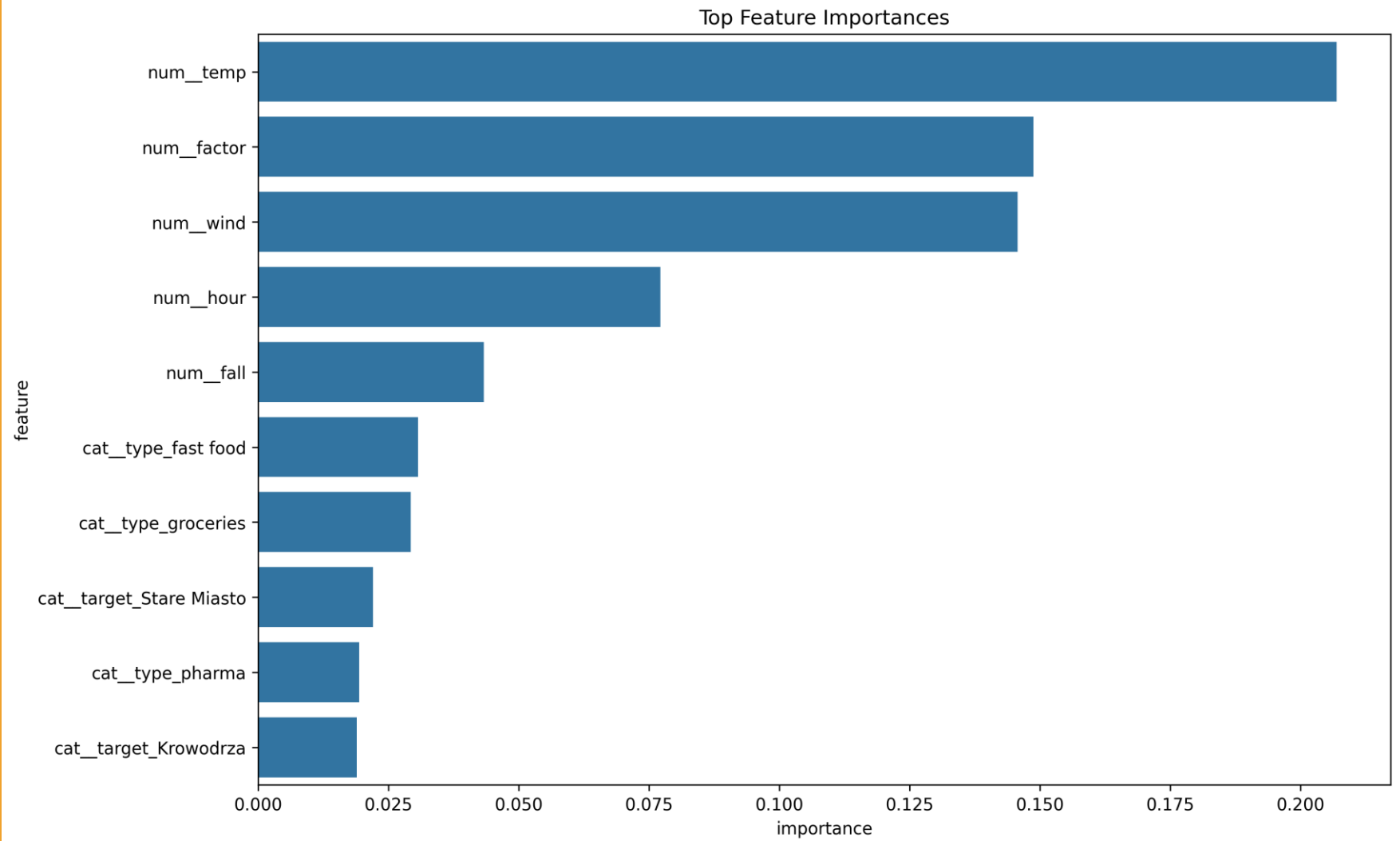
- MAE
- RMSE
- R^2

Evaluations

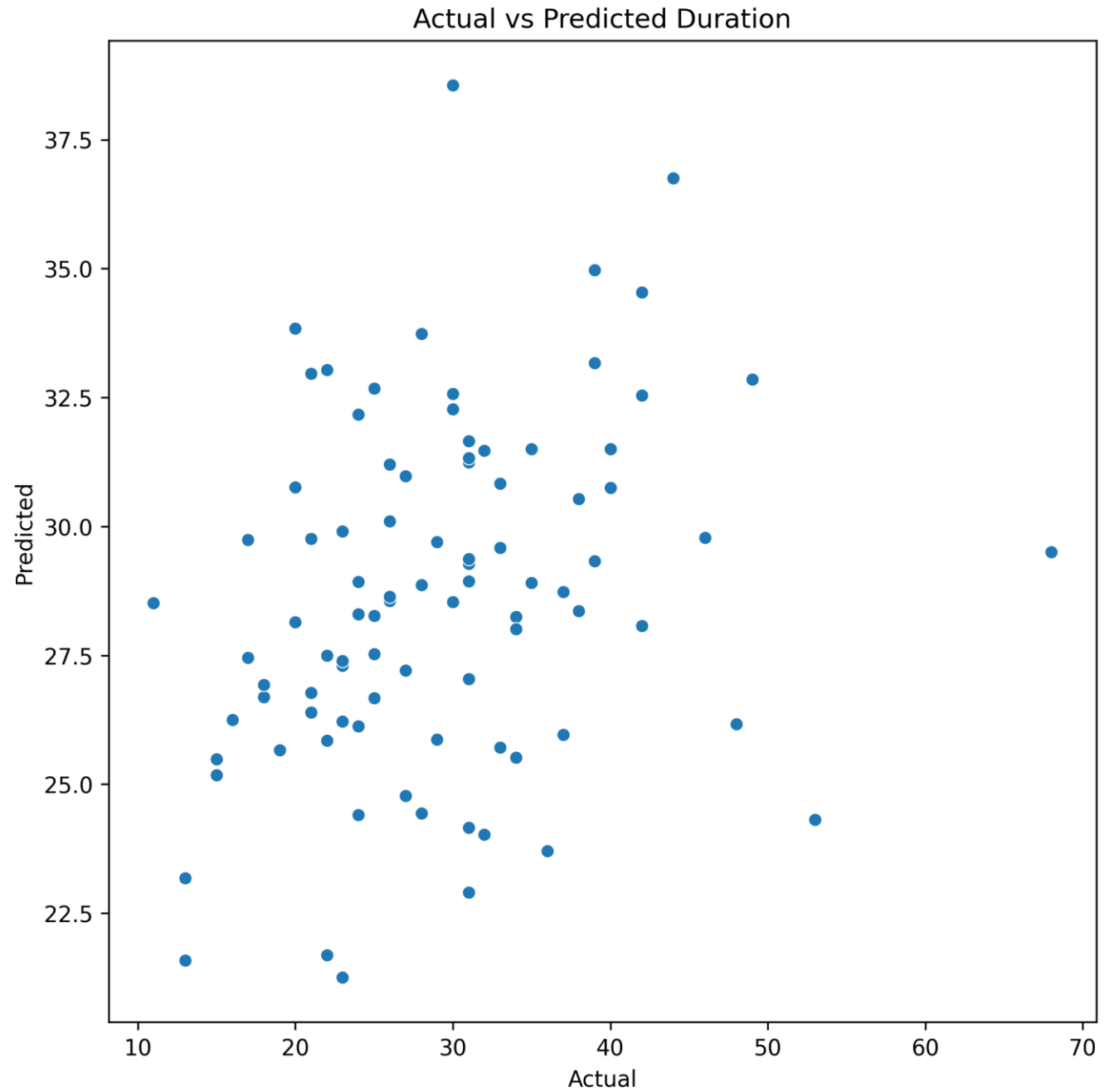
Model	MAE	RMSE	R ²
Random Forest	7.01	9.18	0.08
Linear Regression	7.61	9.94	-0.08

- **Interpretation:**
- Random Forest outperformed Linear Regression.
- Prediction error remained relatively high.
- Low R² suggests substantial operational variability.

Feature Importance



Prediction vs Actual



Business recommendat ions

Operational recommendations:

- Increase courier availability during evening peak hours.
- Introduce weather-related incentive systems.
- Optimize staffing during high-demand periods.
- Monitor district-level operational efficiency.

Future improvements:

- GPS route integration
- Real-time traffic data
- Restaurant waiting time analysis
- Advanced boosting models

See the
interactive
dashboard
in PowerBI!



Courier Data Analysis

The operational & financial dashboard

Content

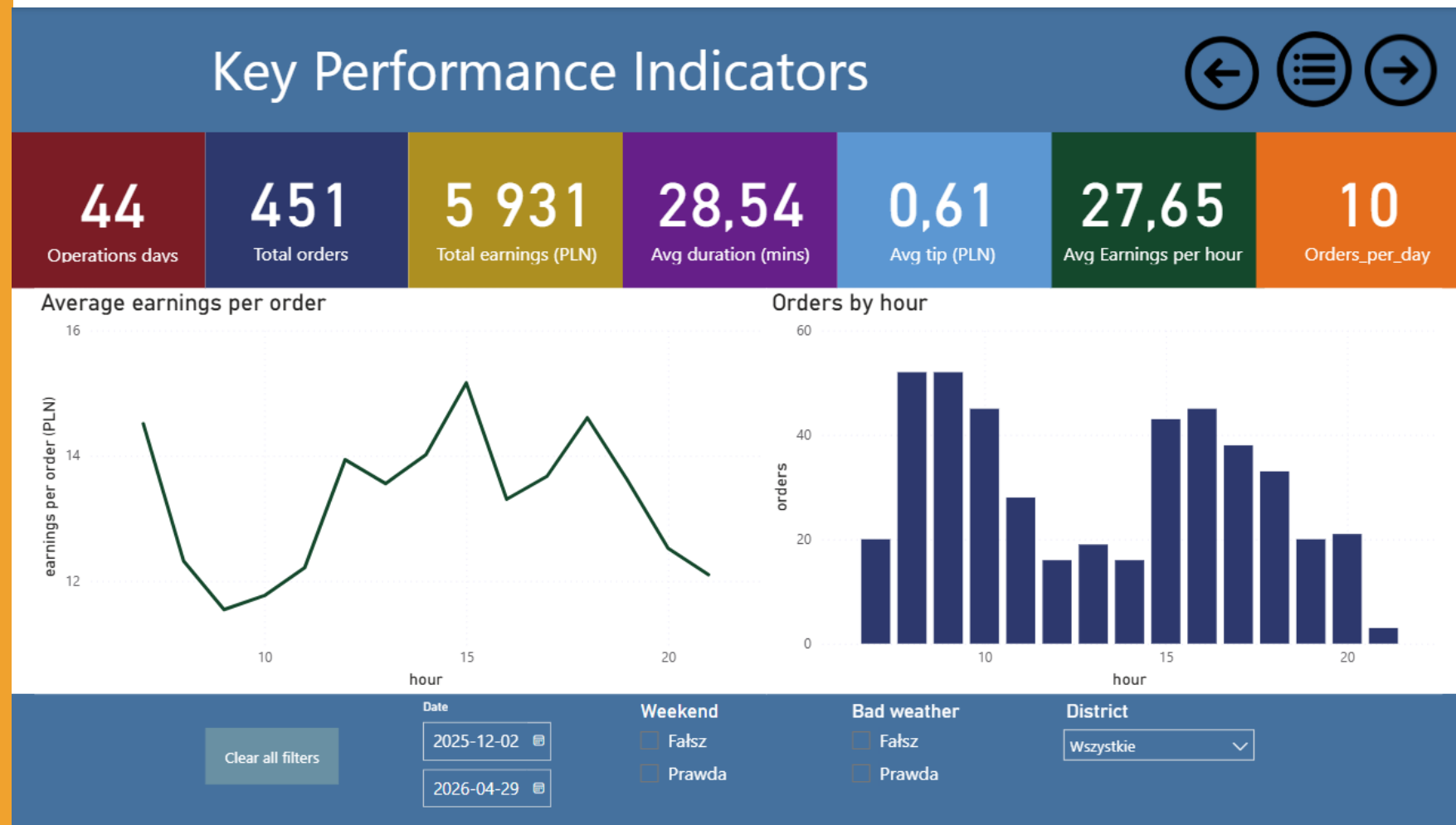
KPI

Operations

ML

Insights

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Conclusions

Final takeaways:

- Delivery operations strongly depend on time and weather conditions.
- Evening hours are the most profitable operational period.
- Machine learning models showed moderate predictive performance.
- Real-world delivery operations contain significant variability.

Skills demonstrated

- Data Cleaning
- Exploratory Data Analysis
- Statistical Analysis
- Machine Learning
- Business Intelligence
- Data Visualization

Thank You

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